

5ME42A(ME) – Manufacturing Technology

5th Semester B.E. Mechanical Engineering

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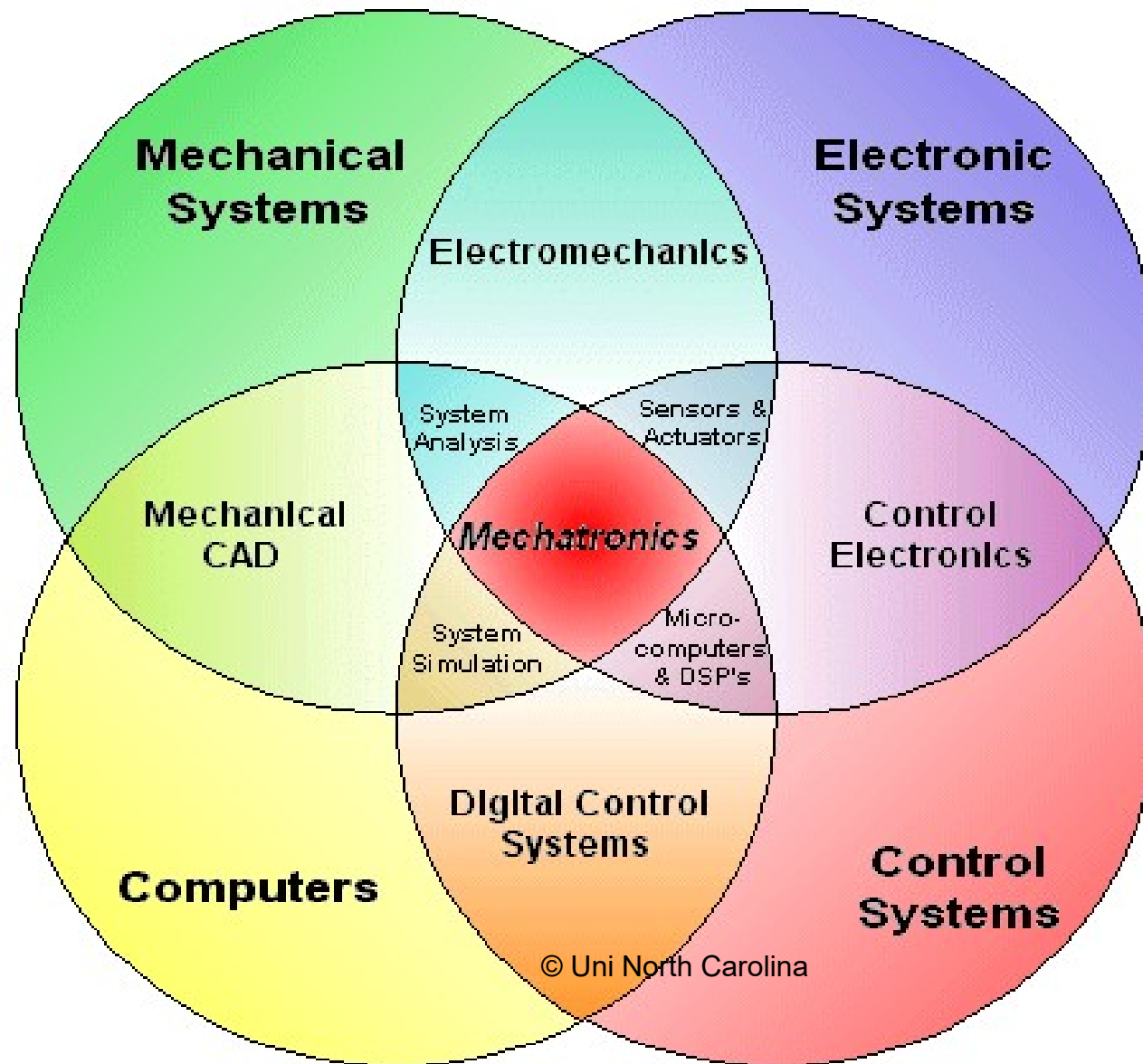
MBM University Jodhpur

What is the Mechatronics?

Mechatronics basically refers to **mechanical electrical** systems and is centered on **mechanics, electronics, computing** and **control** which, combined, make possible the generation of simpler, more economical, reliable and versatile systems.

The term "mechatronics" was first assigned by Mr. Tetsuro Mori, a senior engineer of the Japanese company Yaskawa, in 1969.

What is the Mechatronics?





Common Examples of Mechatronic Devices

Domestic

- microwaves, breadmakers, washing machines, dishwashers, VCRs, CD players ...

Industrial

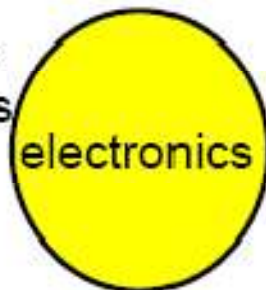
- CNC machine tools, active magnetic bearings, active noise suppression systems, mobile robots ...

Automotive

- Antilock brake systems, fuel/ ignition/emission controls, climate control, cruise control ...

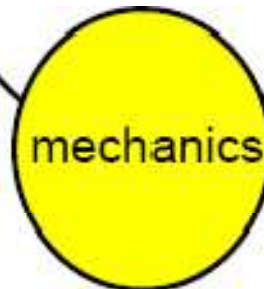
Adding a microprocessor to a device can increase its functionality, but also introduces a new failure mode.

micro electronics
power electronics
sensors
actuators



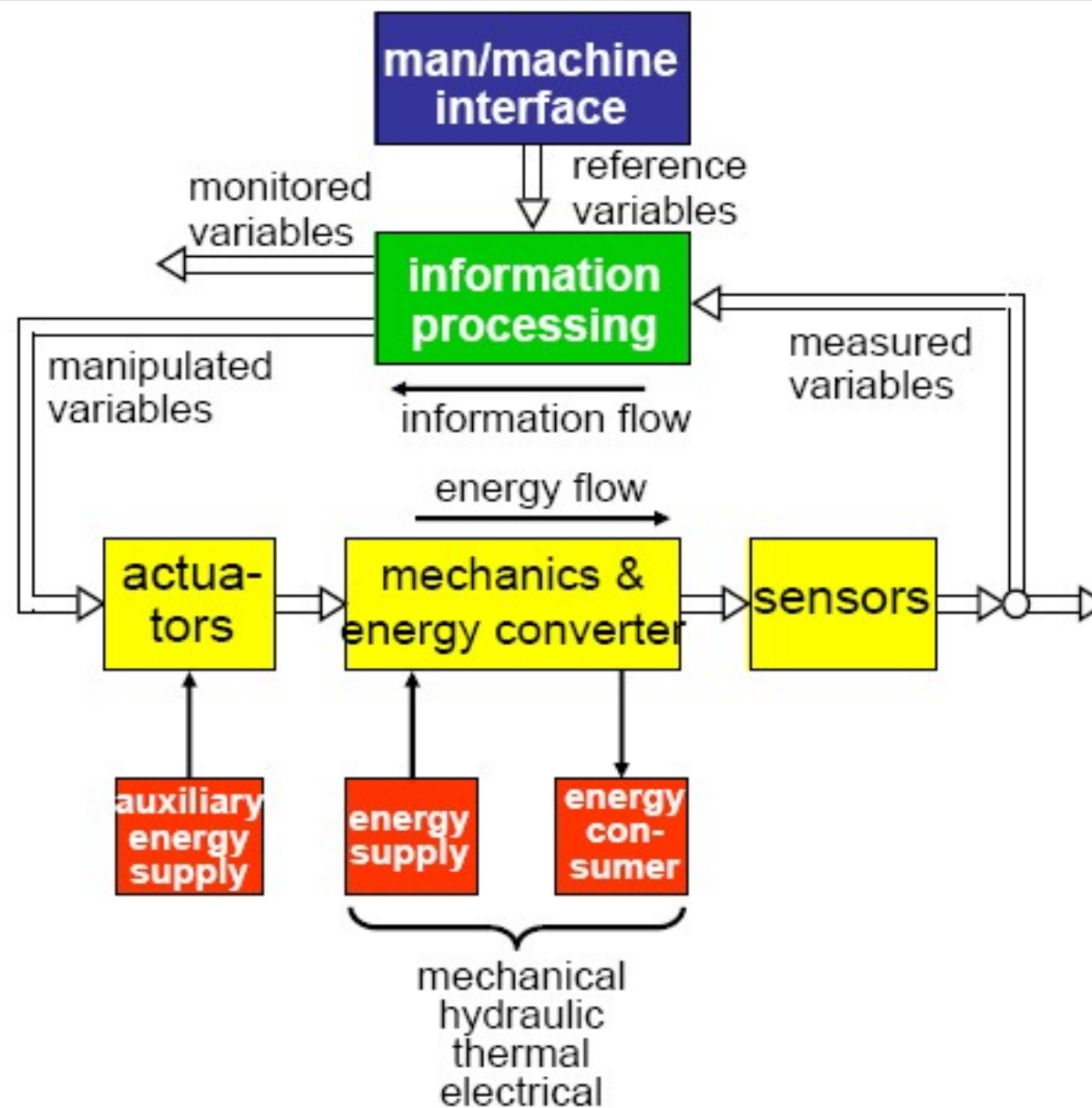
system theory
modeling
automation technology
software
computational intelligence

MECHATRONICS



mechanical elements
machines
precision mechanics

Mechatronics: Synergetic integration of different disciplines



Mechanical process and information processing develop towards mechatronic systems

Mechatronic Systems

Mechatronic machine components

- semi-active hydraulic dampers
- automatic gears
- magnetic bearings



Mechatronic motion generators

- integrated electrical servo drives
- integrated hydraulic servo drives
- integrated pneumatic servo drives
- robots (multi-axis, mobile)

Mechatronic power producing machines

- brushless DC motors
- integrated AC drives
- mechatronic combustion engines



Mechatronic power consuming machines

- integrated multi-axis machine tools
- integrated hydraulic pumps



Mechatronic automobiles

- antilock brake (ABS)
- electro-hydraulic brake (EHB)
- active suspension
- active front steering



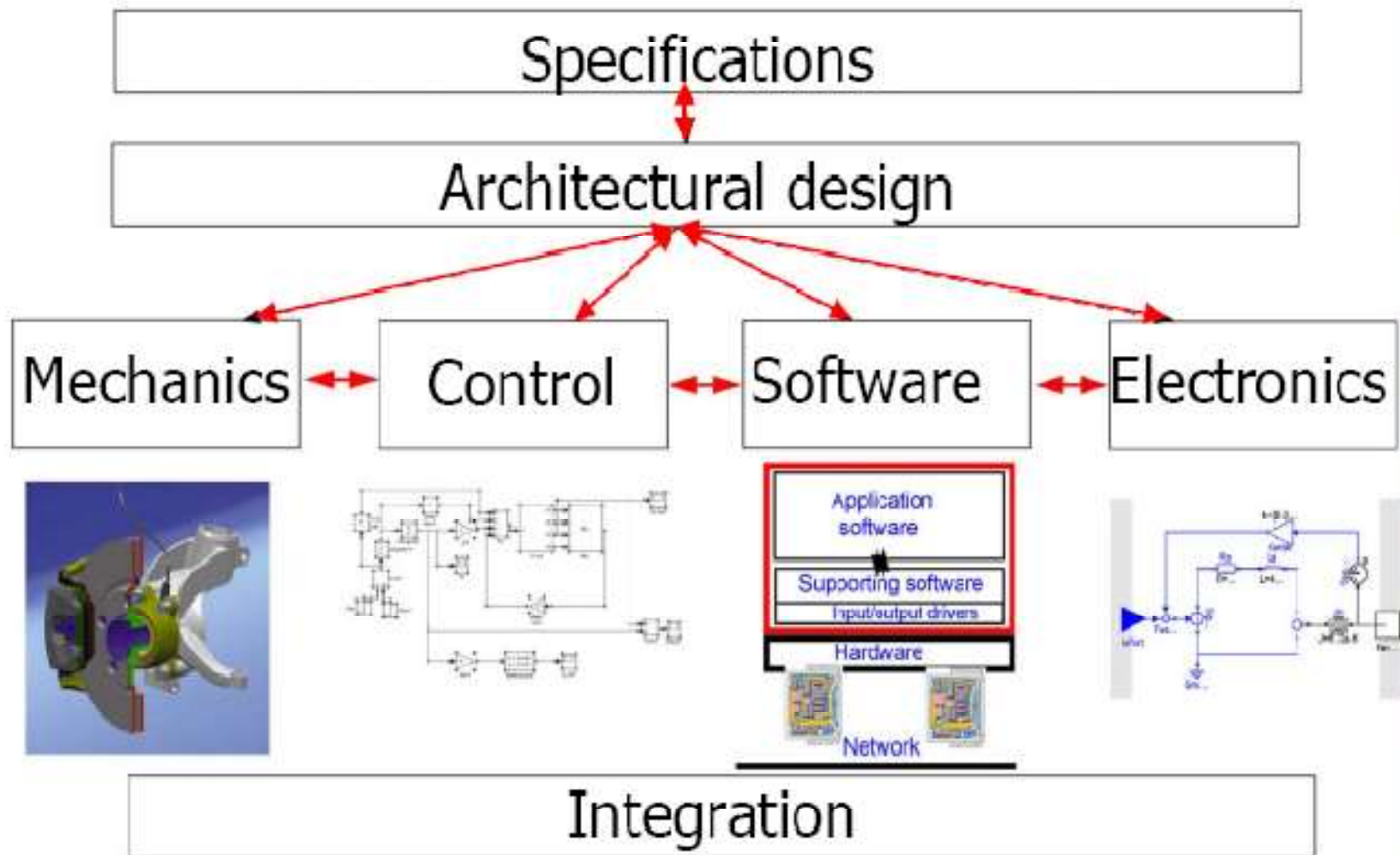
Mechatronic trains

- tilting trains
- active bogie
- magnetic levitated trains (MAGLEV)



Examples for mechatronic systems

Mechatronics engineering



■ Embedded Systems

A combination of **hardware** and **software** which together form a component of a **Mechatronics** systems.

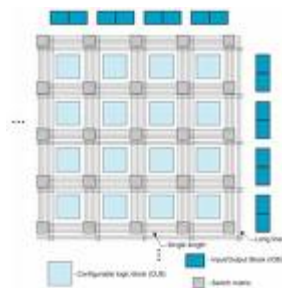
An embedded system is designed to run on its own **without human** intervention, and may be required to respond to events in real time.

Hardware, Software, and Firmware

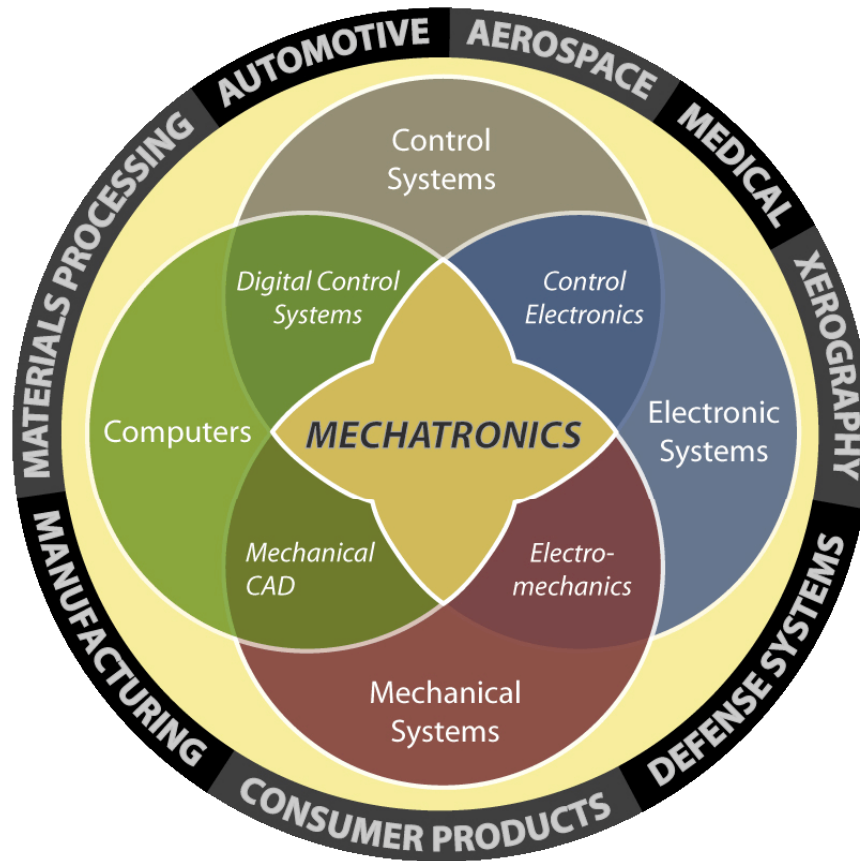
Hardware is the name given to the physical devices and circuitry of the computer.

Software refers to the programs written for the computer.

Firmware is the term given to programs stored in ROMs or in Programmable devices which permanently keep their stored information.



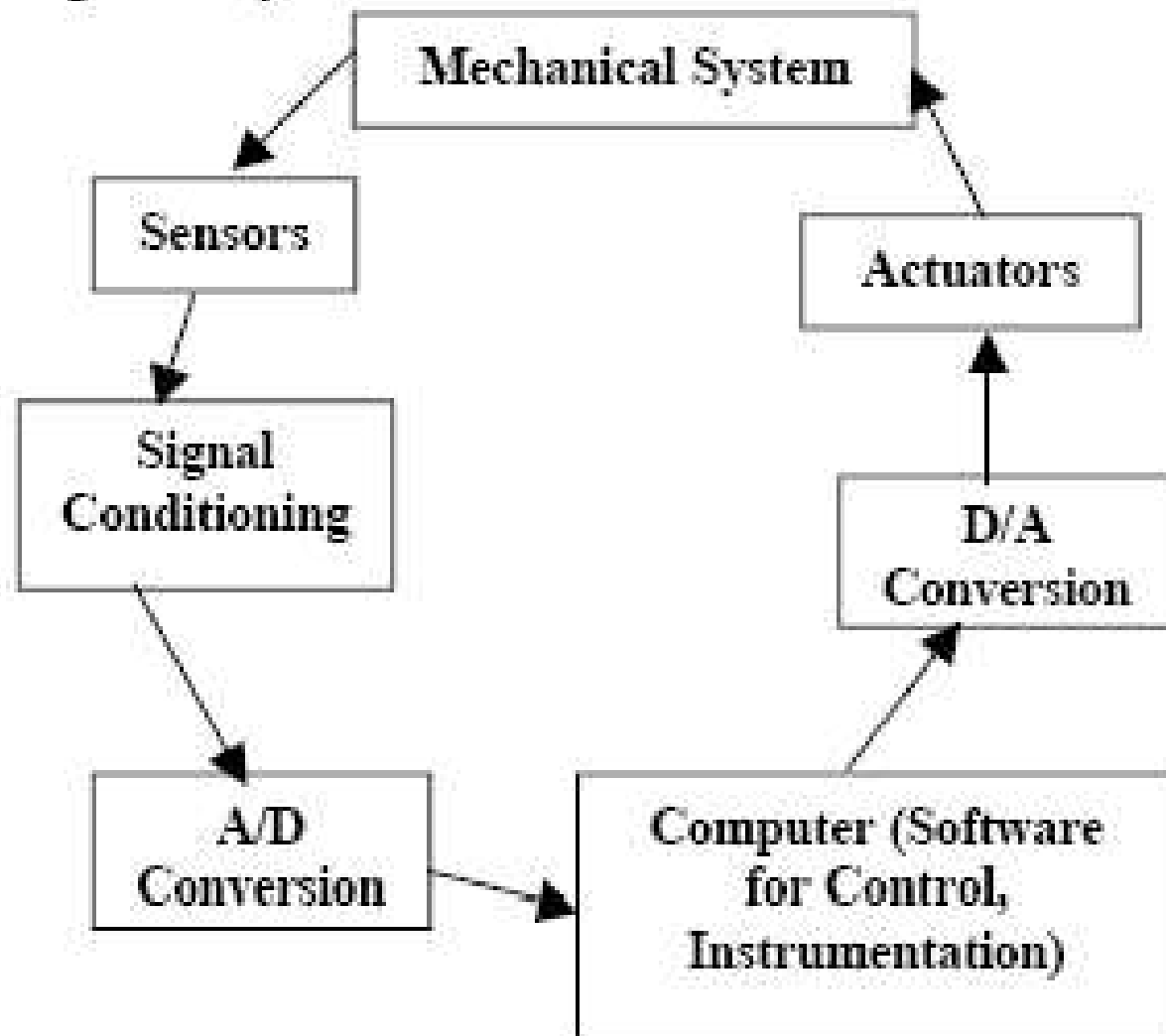
Mechatronics



The synergistic combination of mechanical, electrical, and computer engineering

- Emphasis on integrated design for products
- Optimal combination of appropriate technologies

Mechatronic System Model



Fundamental mechatronic Problems

- Structural integration of mechanical, electronic and information departments into a uniform creative staff;
- Education and training of engineers specialized in mechatronics.
- Integration of information technologies from various scientific and technical fields into a uniform toolkit to provide computer support of mechatronic problems;

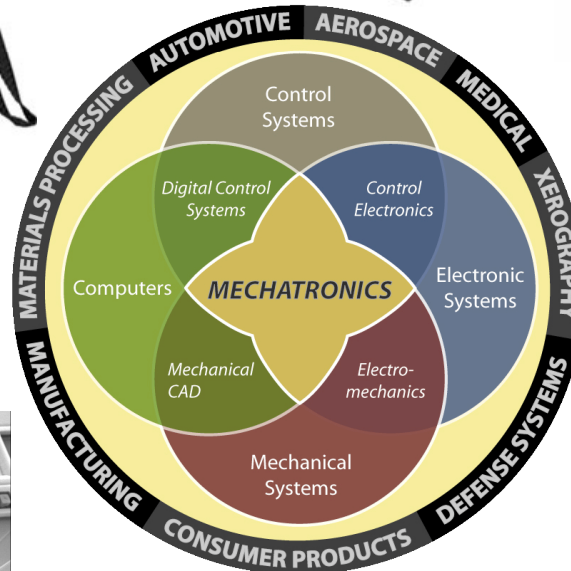
Mechatronics Systems



Consumer Electronics



MEMS



High Speed Trains



Micro to Macro Applications

Mechatronics Systems

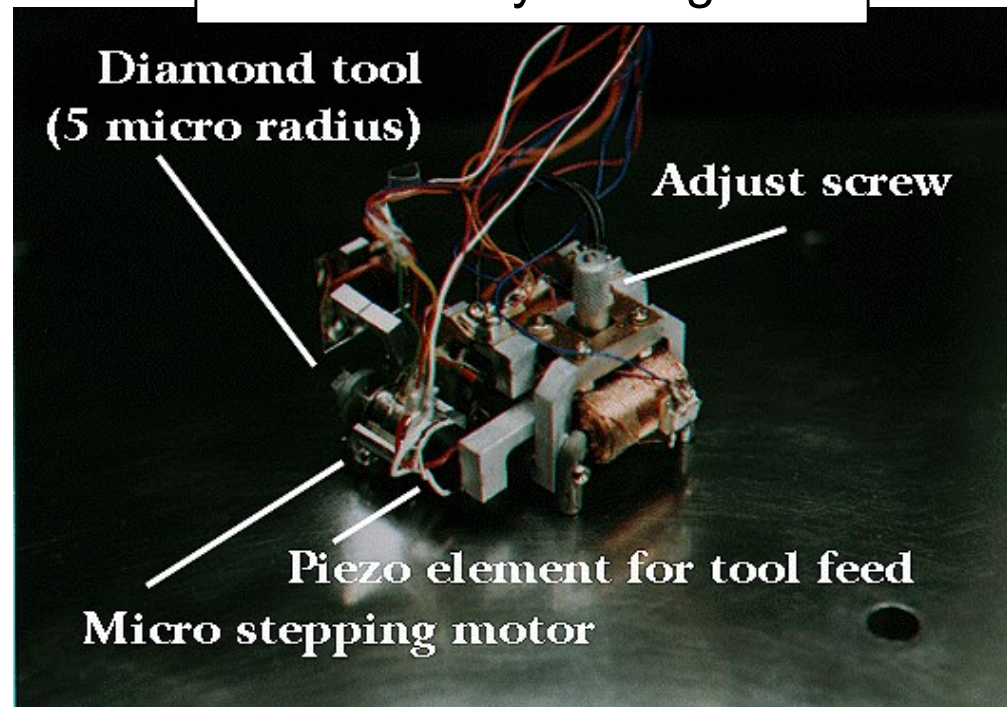
-Manufacturing Applications-

Micro Factory

- Desktop sized Factory
- Build small parts with a small factory
- Greatly reduces space, energy, and materials



Micro Factory Drilling Unit



Mechatronics Systems

-Manufacturing Applications-

CNC Bending

- Fully automated bending: load sheet metal and the finished bent parts come out
- Can bend complex shapes



Mechatronics Systems

-Manufacturing Applications-

CNC Machining

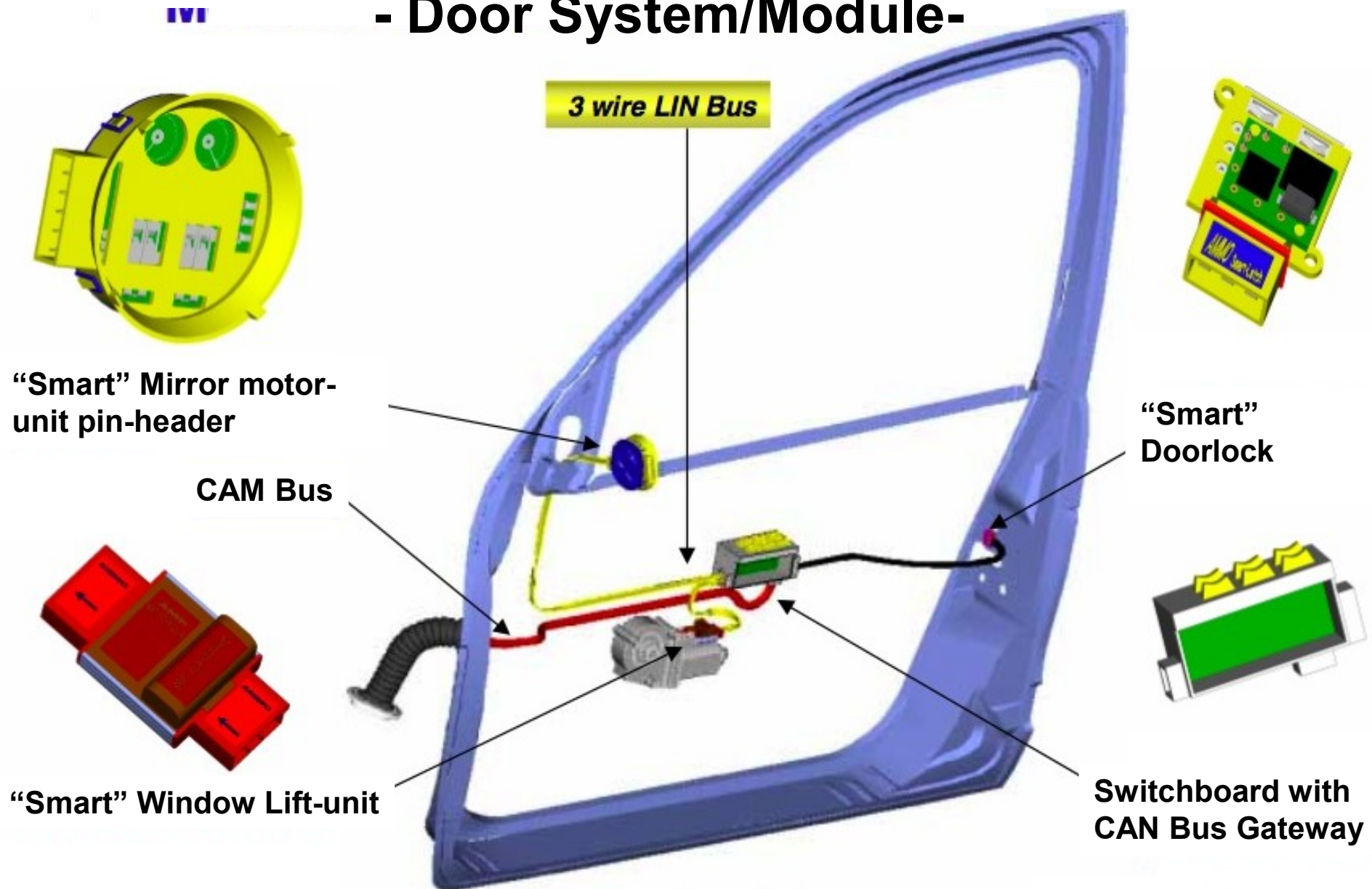


Advantages

- Deliver the highest accuracies
- Can create very complex shapes

Mechatronics Systems

- Door System/Module-



Mechatronics Systems

-Transportation Applications-

High Speed Trains

- Train Position and Velocity constantly monitored from main command center.
- Error margin in scheduling no more than 30 seconds
- Fastest trains use magnetic levitation

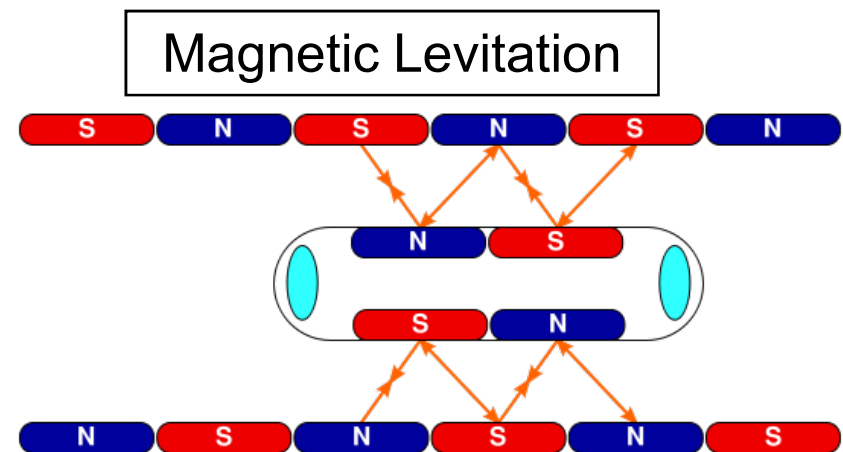
JR-Maglev

Top Speed: 574 km/h (357 mph)

Country: Japan



Transrapid
Top Speed: 550 km/h (340 mph)
Country: German



Mechatronics Systems

-Transportation Applications-

Systems Uses

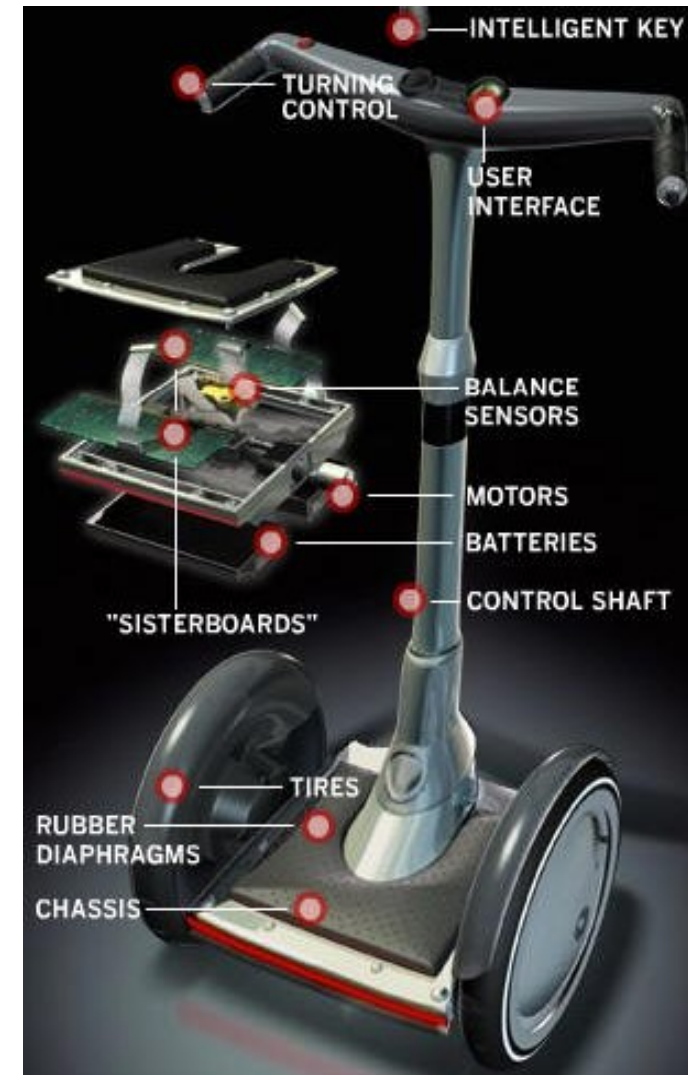
- Tilt and pressure sensors
- Microcontroller
- Motors
- Onboard power source

Segway



Advantages

- Simple and intuitive** personal transportation device



Mechatronics Systems

-Smart Robotics Application-

BigDog



System Can

- Carry 340 lb
- Run 4 mph
- Climb, run, and walk
- Move over rough terrain



Advantages

- Robot with **rough-terrain mobility** that could carry equipment to remote location.

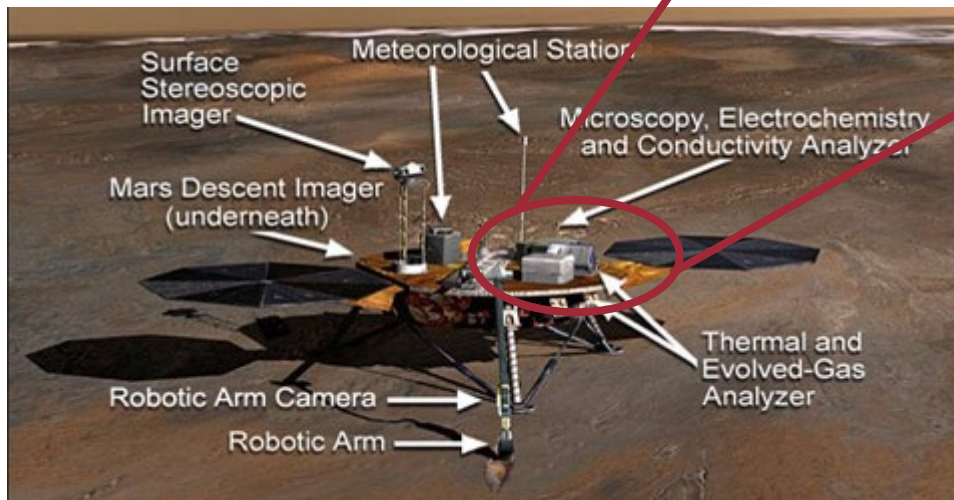
Mechatronics Systems

-Space Exploration Application-

Phoenix Mars Lander's

System Can

- Collect specimens
- Has automated onboard lab for testing specimens



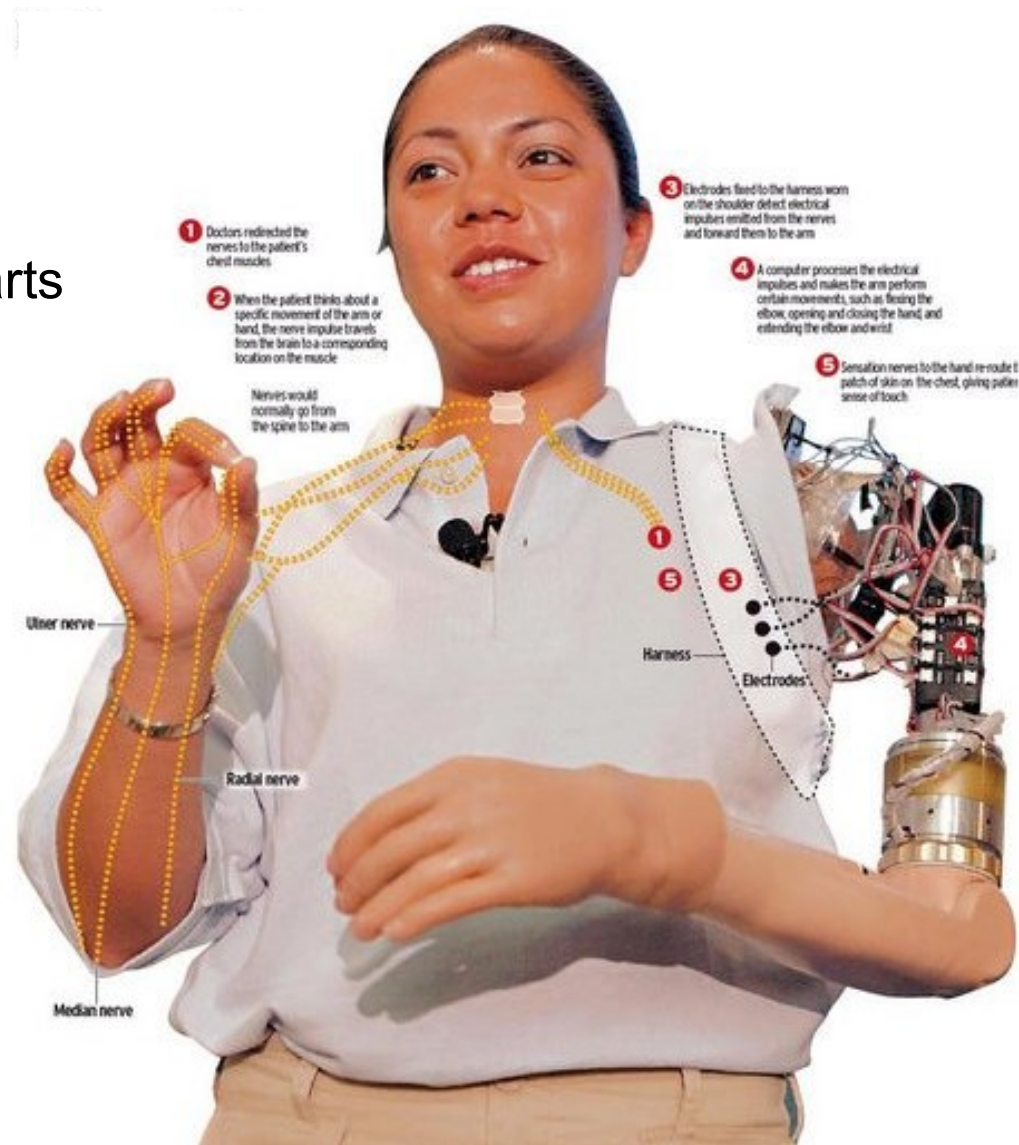
Advantages

- Robot that can travel to other planets and **take measurements automatically.**

Mechatronics Systems -Medical Applications-

Prosthetics

- Arms, Legs, and other body parts can be replaced with electromechanical ones.

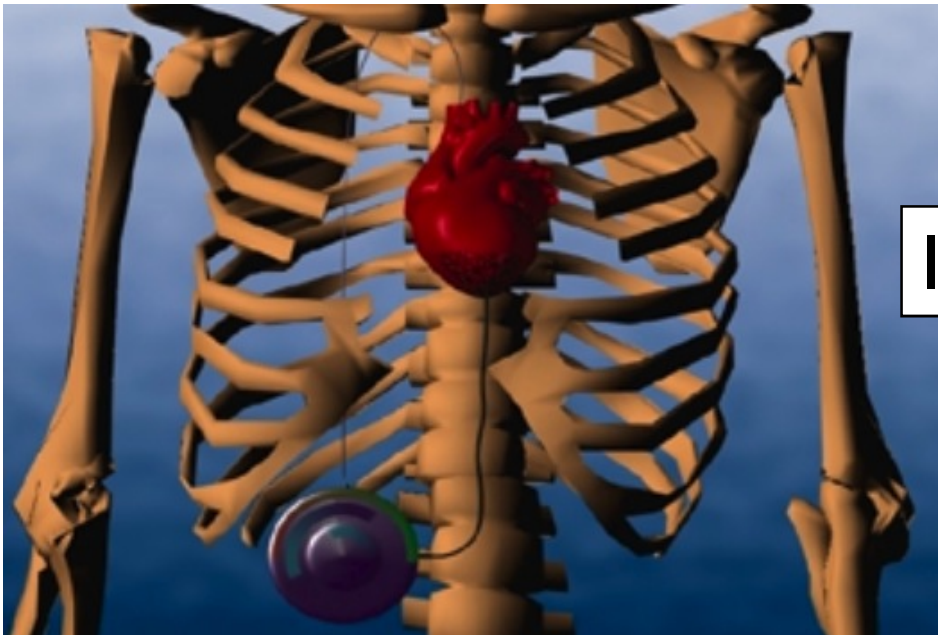


Mechatronics Systems

-Medical Applications-

Pace Maker

- Used by patients with slow or erratic heart rates. The pacemaker will set a normal heart rate when it sees an irregular heart rhythm.



Implantable Defibrillation

- Monitors the heart. If heart fibrillates or stops completely it will shock the heart at high voltage to restore a normal heart rhythm.

Mechatronics Systems

-Defense Applications-

- Advanced technology is making our soldiers safer.
- Some planes can now be flown remotely.

Unmanned Aerial Vehicle



Stealth Bomber

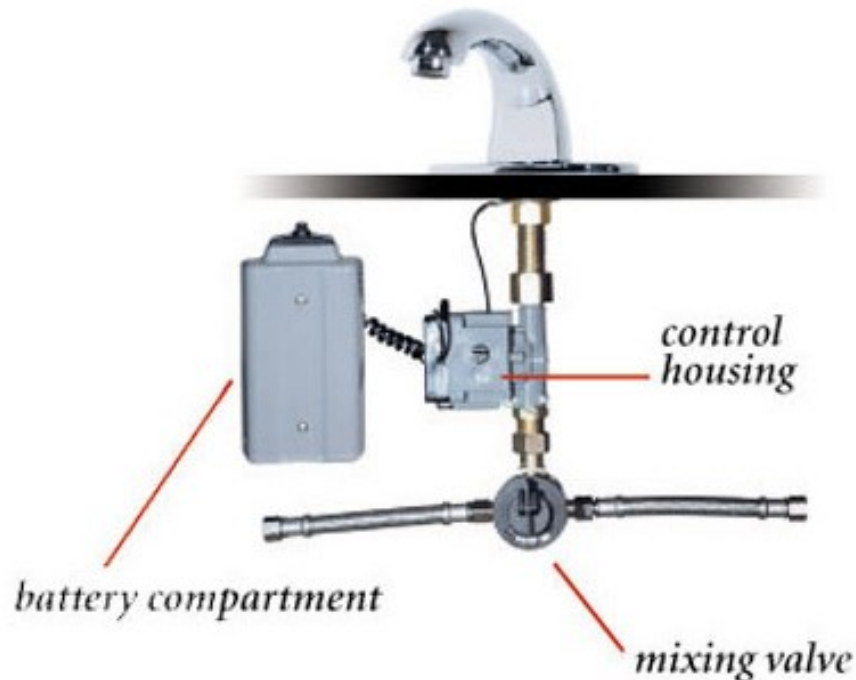


Mechatronics Systems

-Sanitation Applications-

System Uses

- Proximity sensors
- Control circuitry
- Electromechanical valves
- Independent power source



Advantages

- **Reduces spread of germs** by making device hands free
- **Reduces wasted water** by automatically turning off when not in use

Mechatronics Systems

-Sanitation Applications-

Systems Uses

- Motion sensors
- Control circuitry
- Electromechanical actuators
- Independent power source

Soap Dispenser



Paper Towel Dispenser



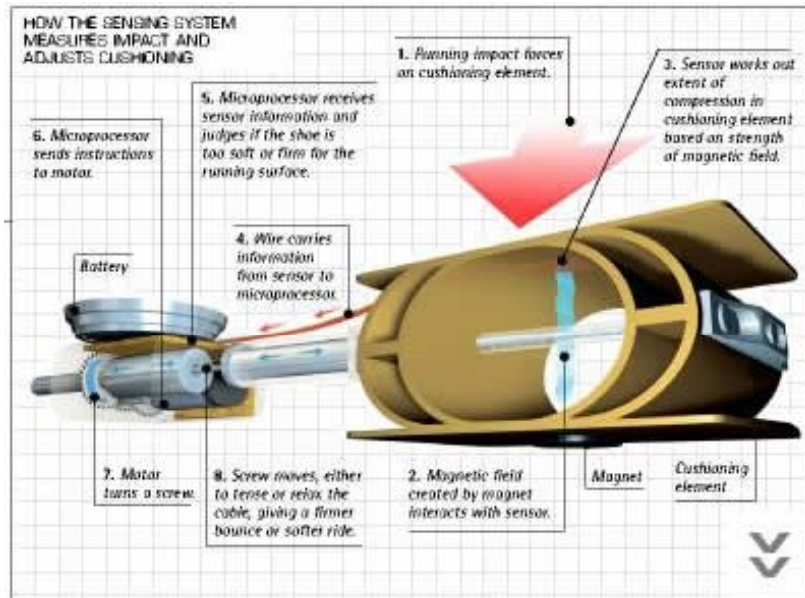
Advantages

- Reduces spread of germs** by making device hands free
- Reduces wasted materials** by controlling how much is dispensed

Mechatronics Systems

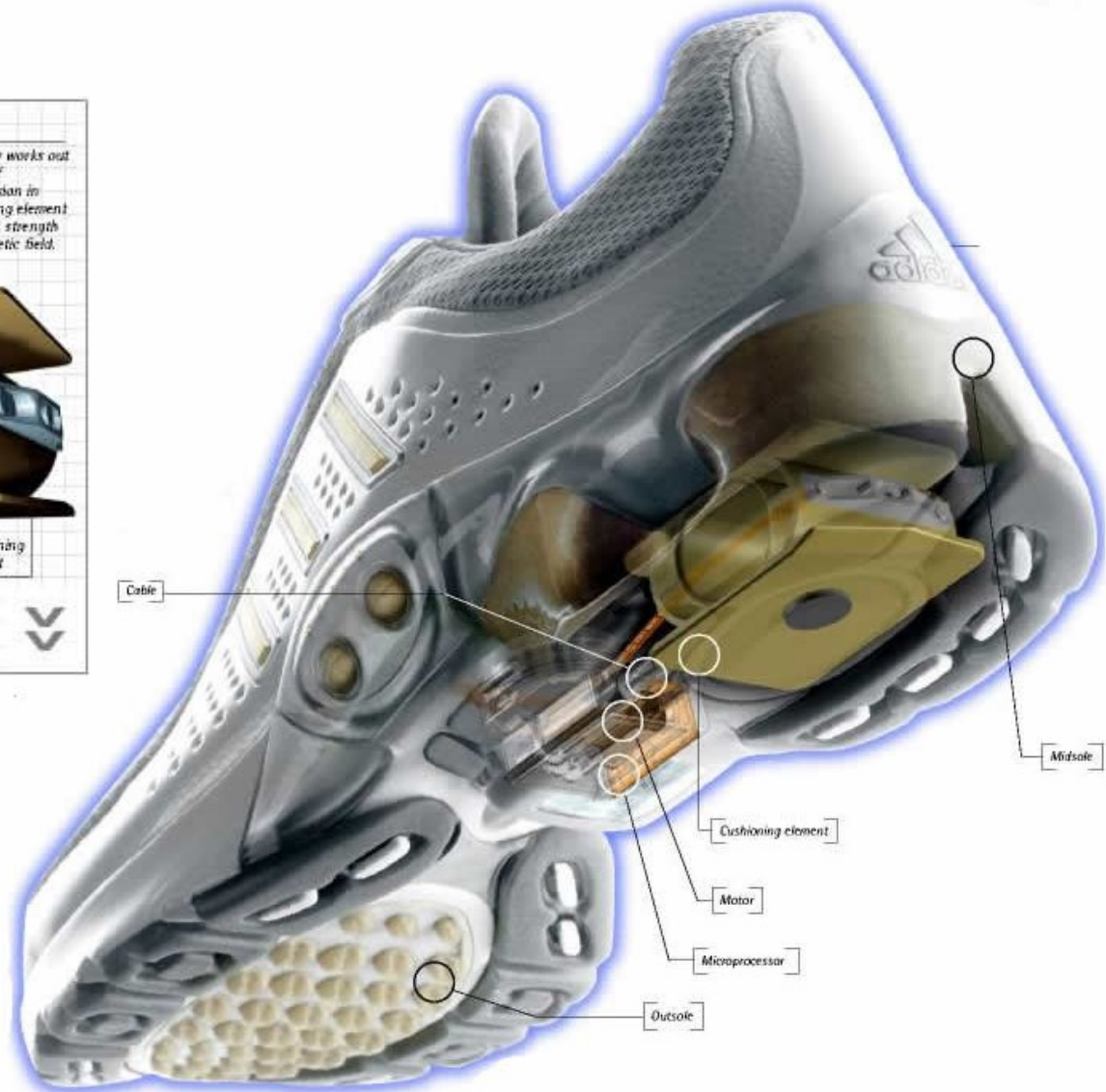
-Sports Applications-

Running Shoes



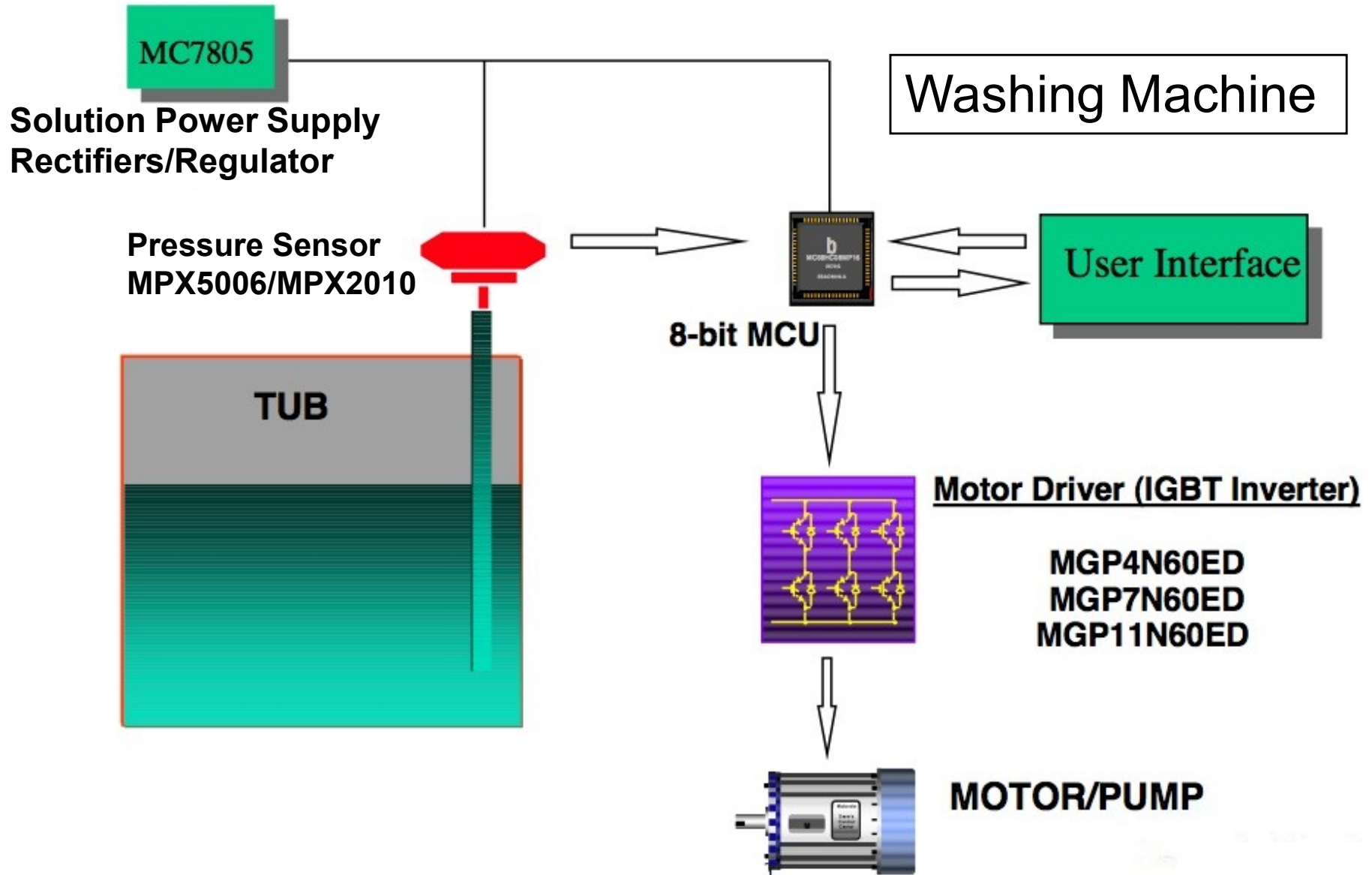
Advantages

- **Automatically changes cushioning** in shoe for different running styles and conditions for improved comfort



Mechatronics Systems

-Smart Home Applications-



Mechatronics Systems

-Smart Home Applications-

